

ICT702 Data Visualisation

Visualising Variability

Module 5



I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



Part 2

Distributions for Quantitative Data: The Histogram

Creating a histogram for quantitative data

	A
1	Age at Death (Years)
2	83
3	76
4	78
5	74
6	35
7	78
8	73
9	84
10	55
11	73

Three features need to be defined:

1. The number of nonoverlapping bins
2. The width (numerical range) for each bin
3. The range spanned by the set of bins

Build a Histogram: Definitions and Application

Definitions

Number of bins: between 5 and 20, depending on the number of observations.

Width of the bins: same for all bins.

Approximate Bin Width =
$$\frac{\text{largest value} - \text{smallest value}}{\text{number of bins}}$$

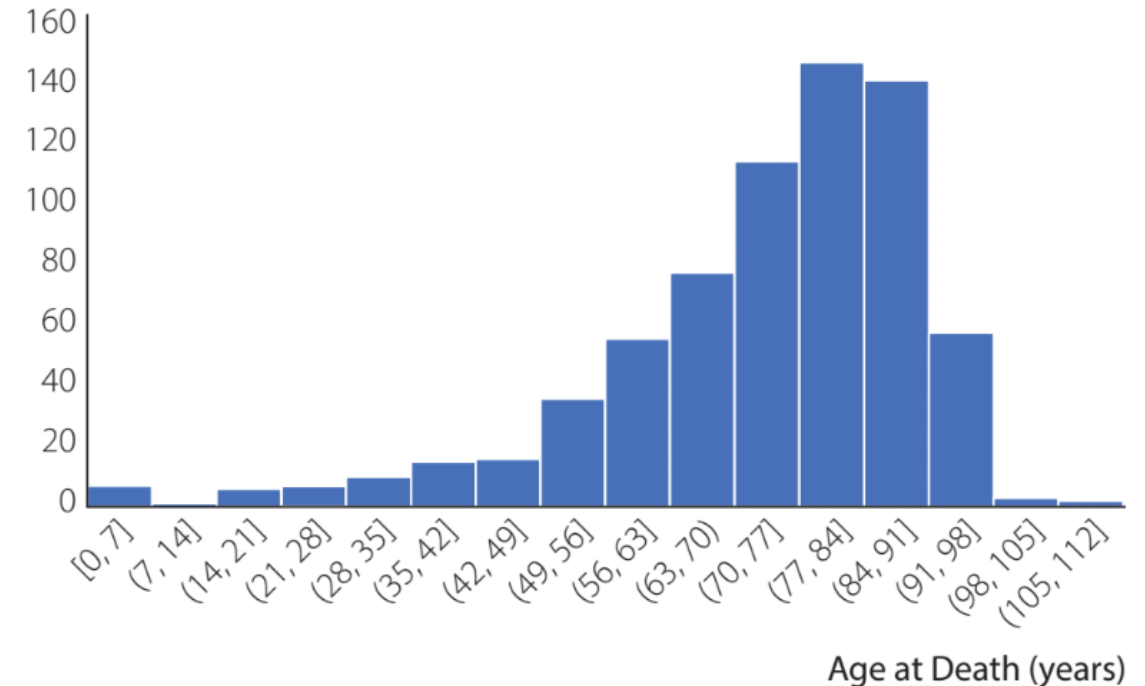
Range spanned by bins: the first bin should begin at a value such that it includes the smallest value in the data.

Histogram for the Data from the Death file

	A	B	C	D	E	F
1	Age at Death (Years)	Bin Lower Limit	Bin Upper Limit	Frequency		Labels
2	83	0	7	7		[0, 7]
3	76	7	14	1		(7, 14]
4	78	14	21	6		(14, 21]
5	74	21	28	7		(21, 28]
6	35	28	35	10		(28, 35]
7	78	35	42	15		(35, 42]
8	73	42	49	16		(42, 49]
9	84	49	56	36		(49, 56]
10	55	56	63	56		(56, 63]
11	73	63	70	78		(63, 70]
12	35	70	77	115		(70, 77]
13	78	77	84	148		(77, 84]
14	65					
15	81					
16	109					
17	91					
18	87					
19	76					

	A	B	C	D	E	F
1	Age at Death (Years)	Bin Lower Limit	Bin Upper Limit	Frequency		Labels
2	83	0	7	=FREQUENCY(A2:A701,D2:D17)	=CONCAT("["	C2," ",D2,"]")
3	76	7	14		=CONCAT("["	C3," ",D3,"]")
4	78	14	21		=CONCAT("["	C4," ",D4,"]")
5	74	21	28		=CONCAT("["	C5," ",D5,"]")
6	35	28	35		=CONCAT("["	C6," ",D6,"]")
7	78	35	42		=CONCAT("["	C7," ",D7,"]")
8	73	42	49		=CONCAT("["	C8," ",D8,"]")
9	84	49	56		=CONCAT("["	C9," ",D9,"]")
10	55	56	63		=CONCAT("["	C10," ",D10,"]")
11	73	63	70		=CONCAT("["	C11," ",D11,"]")
12	35	70	77		=CONCAT("["	C12," ",D12,"]")
13	78	77	84		=CONCAT("["	C13," ",D13,"]")
14	65	84	91		=CONCAT("["	C14," ",D14,"]")
15	81	91	98		=CONCAT("["	C15," ",D15,"]")
16	109	98	105		=CONCAT("["	C16," ",D16,"]")
17	91	105	112		=CONCAT("["	C17," ",D17,"]")
18	87				=CONCAT("["	D17," ",D17,"]")
19	76					

Distribution of Age at Death

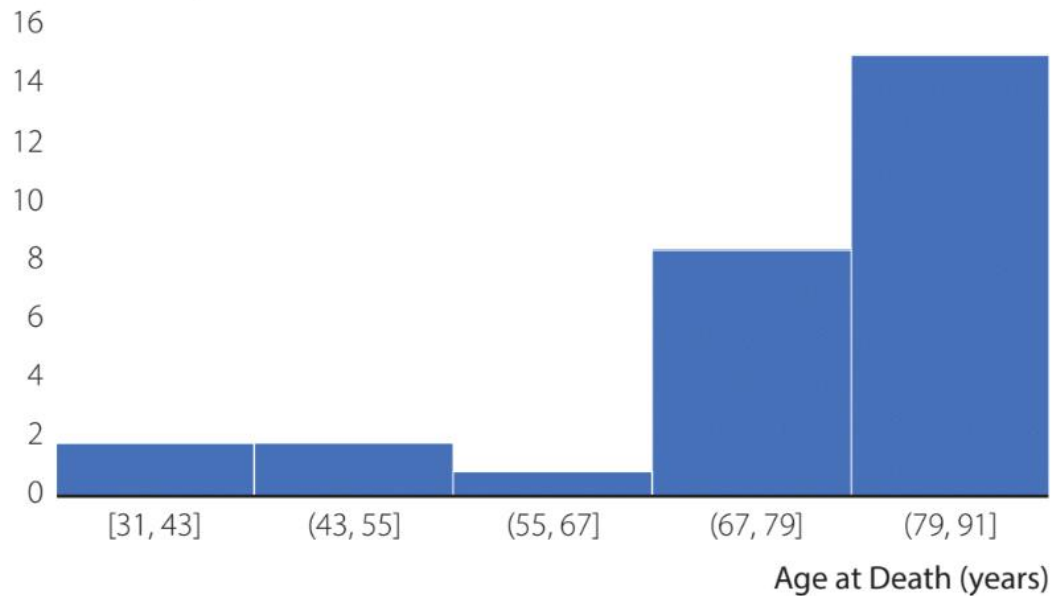


Effect of the Number of Bins on a Histogram

Varying the number of bins in the histogram for the age at death distribution

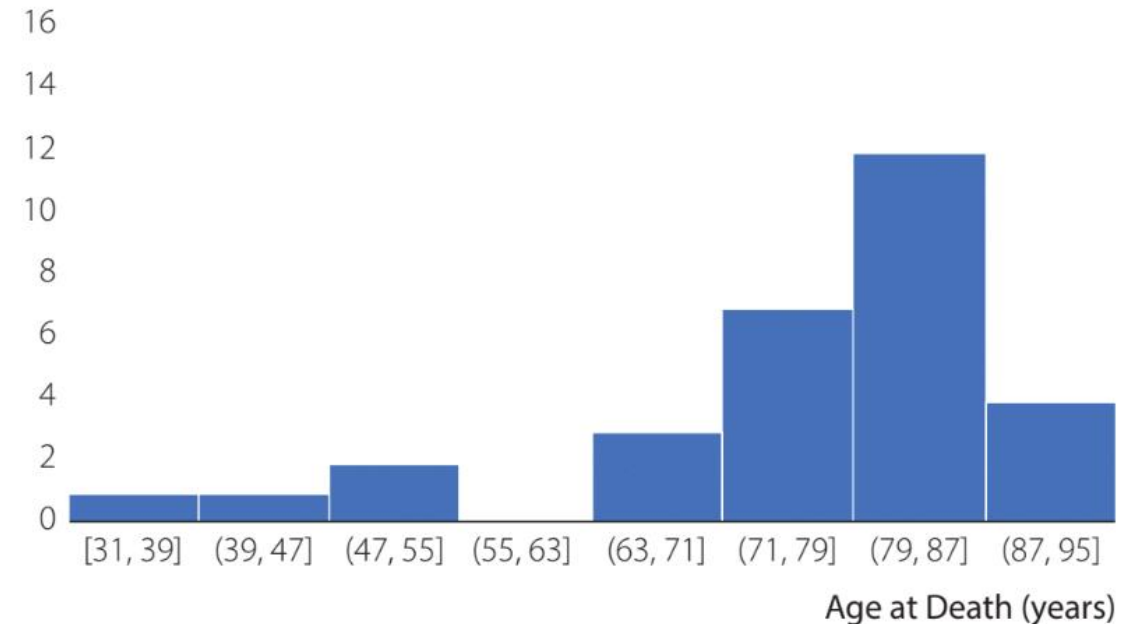
Distribution of Age at Death: 5 Bins

Count of People



Distribution of Age at Death: 8 Bins

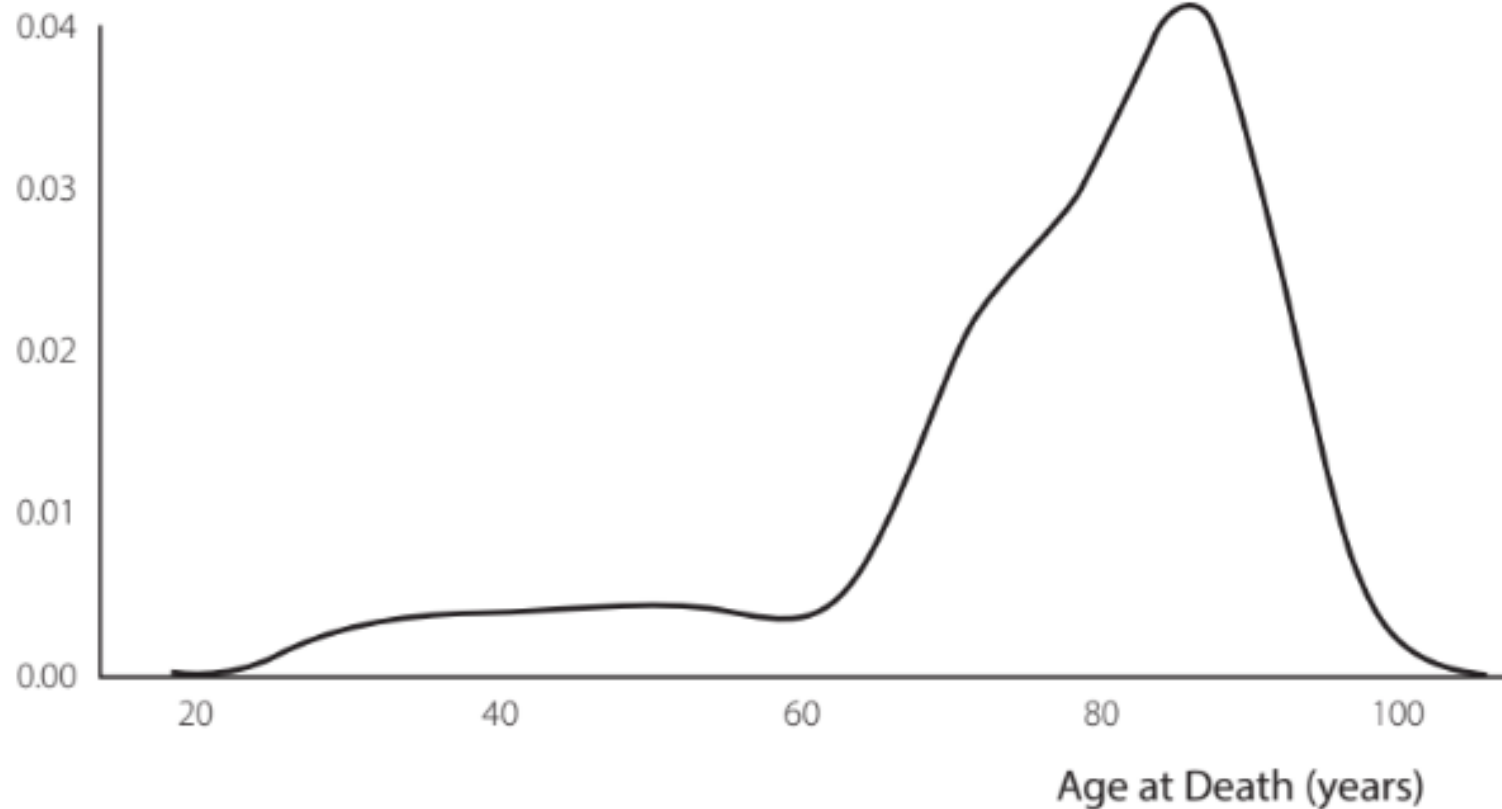
Count of People



Additional Charts: the Kernel Density Chart

Distribution of Age at Death

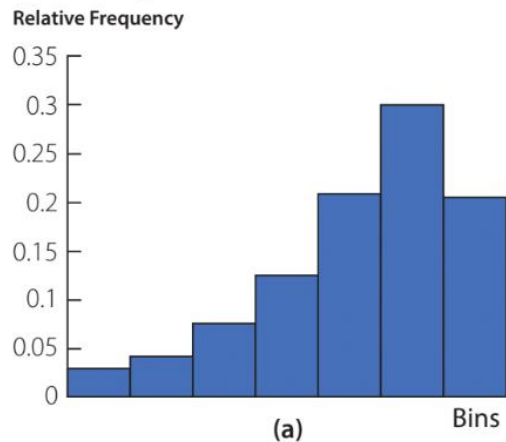
Probability Density



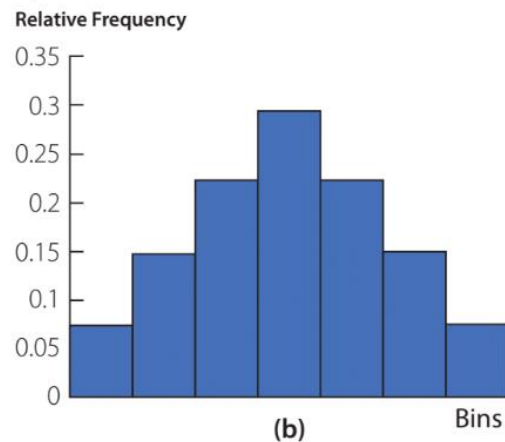
Skewness in a Quantitative Distribution

Skewness represents the lack of symmetry in a quantitative distribution.

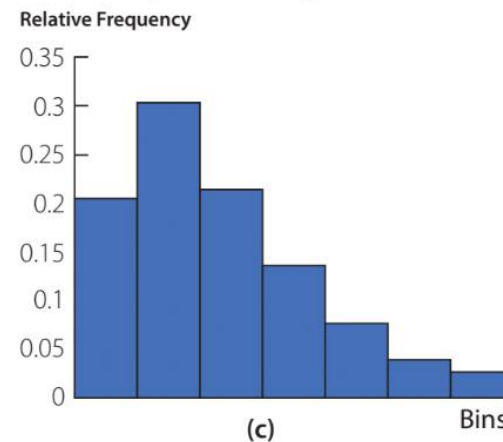
Moderately: Skewed Left



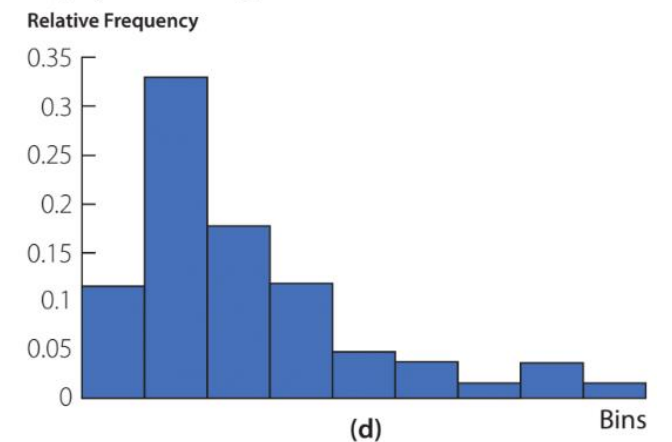
Symmetric



Moderately: Skewed Right



Highly Skewed Right



Comparing Distributions: the Frequency Polygon

A **frequency polygon** is a visualization tool useful for comparing distributions.

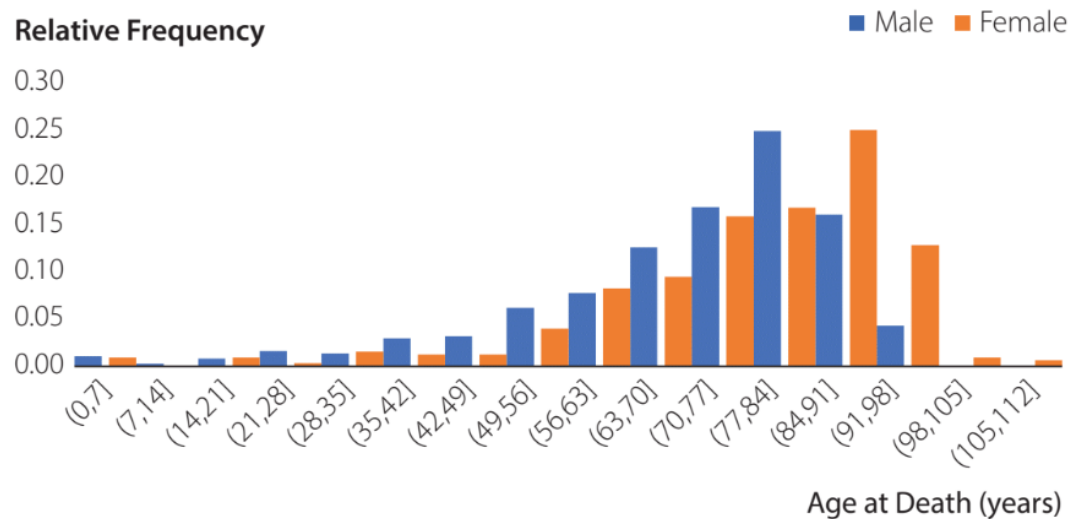
- Like a histogram, a frequency polygon plots the count of observations in a set of bins.
- Unlike a histogram, a frequency polygon uses lines instead of columns to connect the counts of different bins.
- In Excel, use the **Line Chart** option to create a frequency polygon.

Comparing Displays for Quantitative Distributions

Comparing a clustered column display to a frequency polygon display

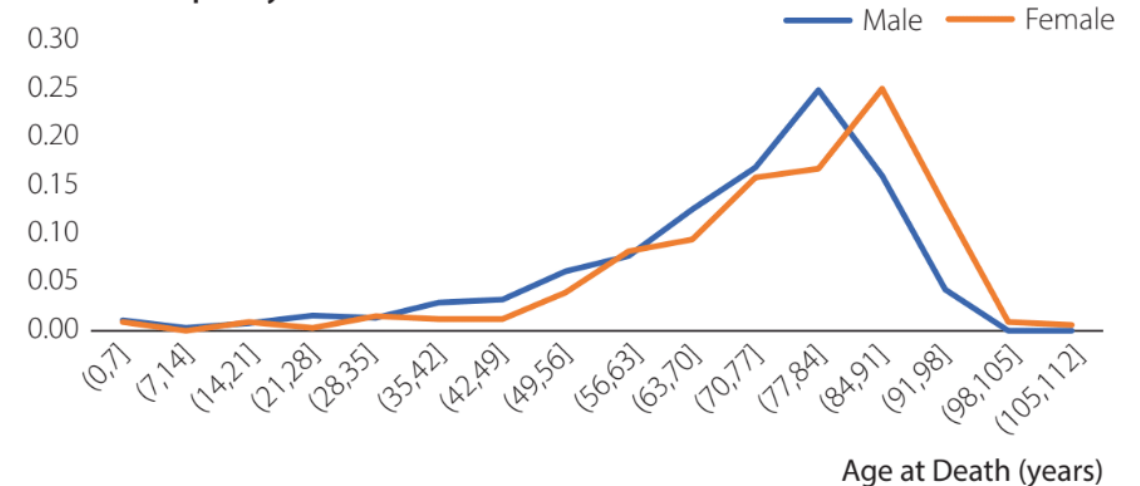
Distribution of Age at Death

Relative Frequency



Distribution of Age at Death

Relative Frequency

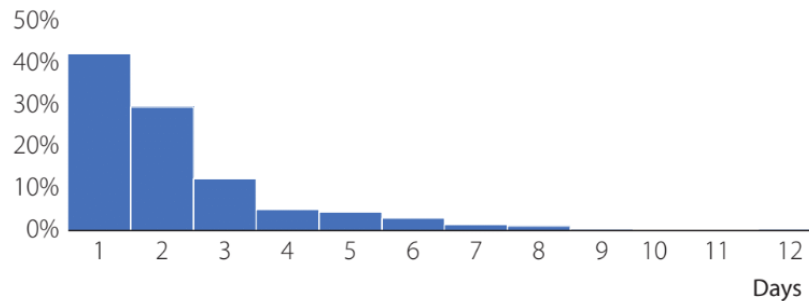


Comparing Quantitative Distributions: Trellis Display

A Trellis display of the length of stay distributions for three hospitals

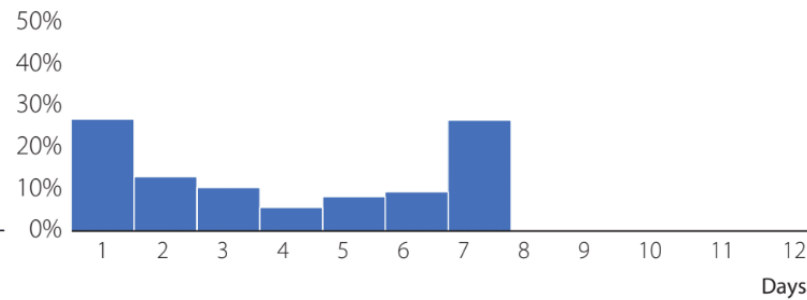
Length of Stay: Seattle Grace

Percent Frequency



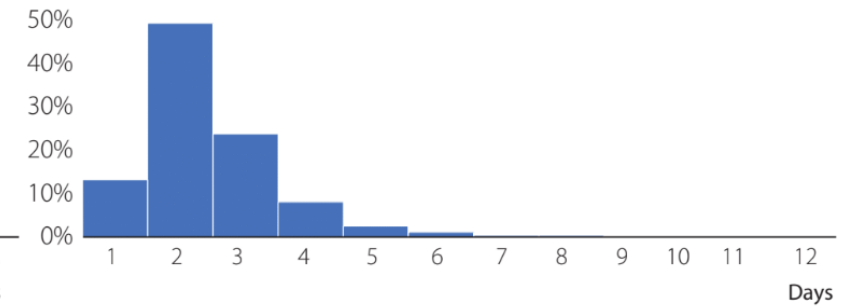
Length of Stay: Seattle Mercy West

Percent Frequency



Length of Stay: Grey Sloan Memorial

Percent Frequency



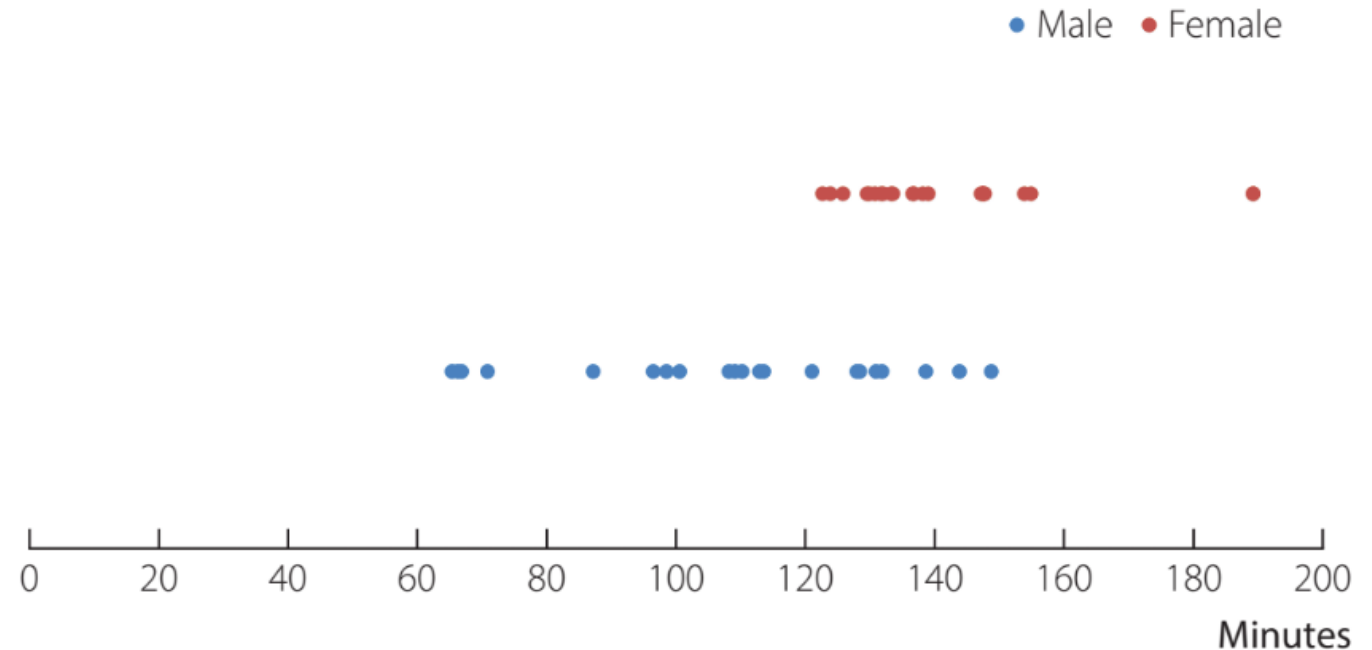
Displaying Individual Values: the Strip Chart

Portion of data in the file *HalfMarathon*

	A	B
1	Sex	Time (Minutes)
2	Male	148.70
3	Female	122.62
4	Male	127.98
5	Female	122.48
6	Female	111.22
7	Male	108.18
8	Female	189.27
9	Male	128.40
10	Female	153.88
11	Female	121.25

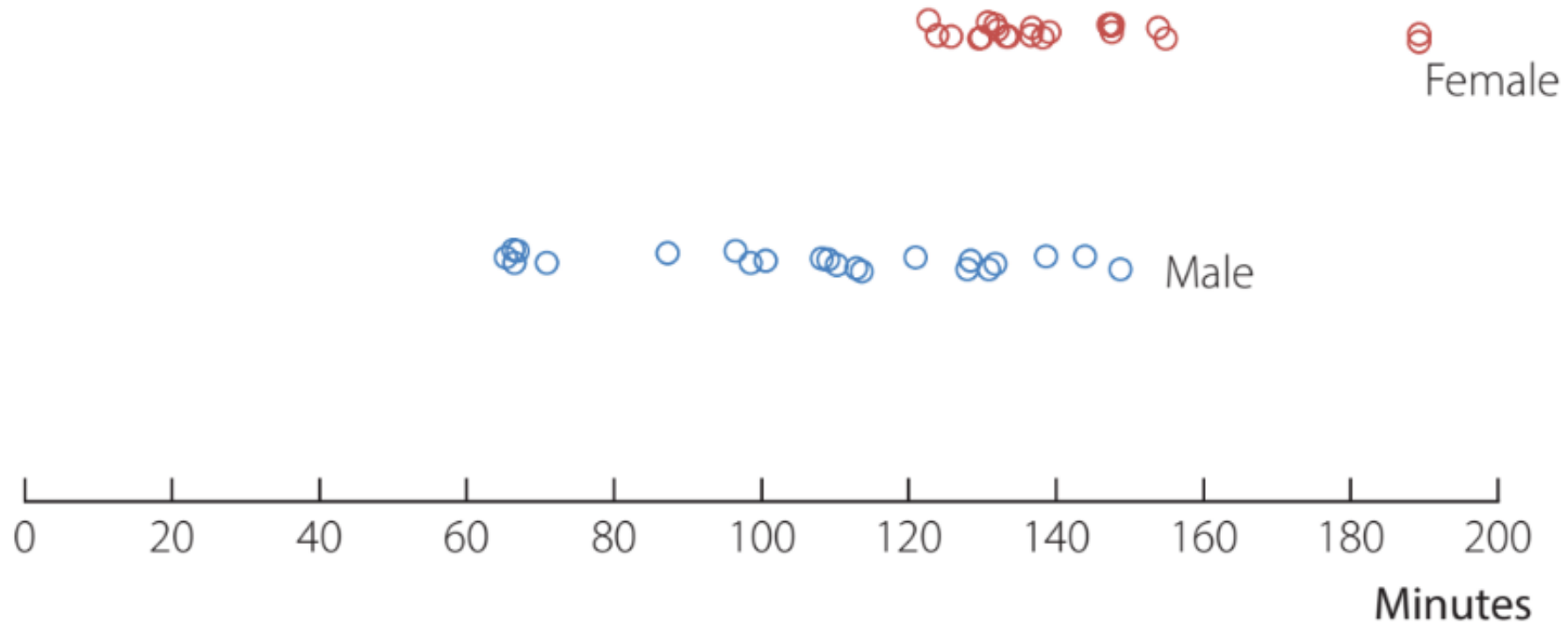
Strip chart of female and male half-marathon race times

Half-Marathon Race Times



A Remedy to Occlusion: the Jittered Strip Chart

Half-Marathon Race Times



End of Part 2

WARNING

This material has been reproduced and communicated to you by or on behalf of the University of the Sunshine Coast in accordance with section 113P of the Copyright Act 1968 (**Act**).

The material in this communication may be subject to copyright under the Act.

Any further reproduction or communication of this material by you may be the subject of copyright protection under the Act.

Do not remove this notice.

